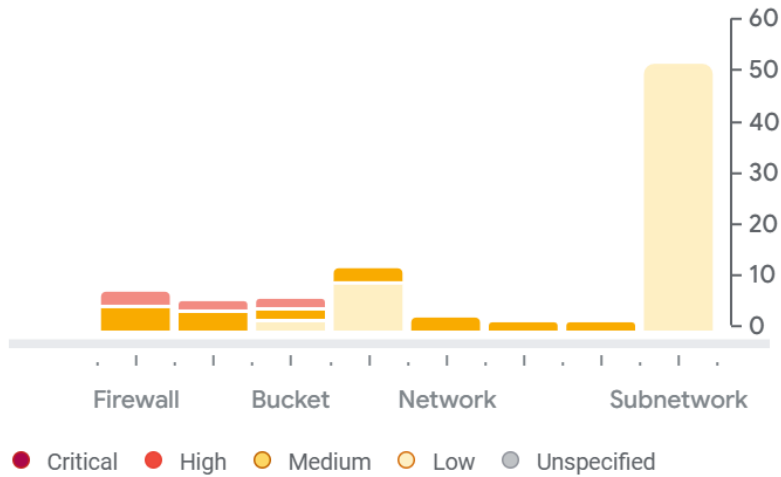


Marc Lavoie

Portfolio Assignment

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Lab Vulnerabilities



Active vulnerabilities

84 active vulnerabilities

< FINDINGS BY CATEGORY FINDINGS BY RESOURCE TYPE >

Filter Categories

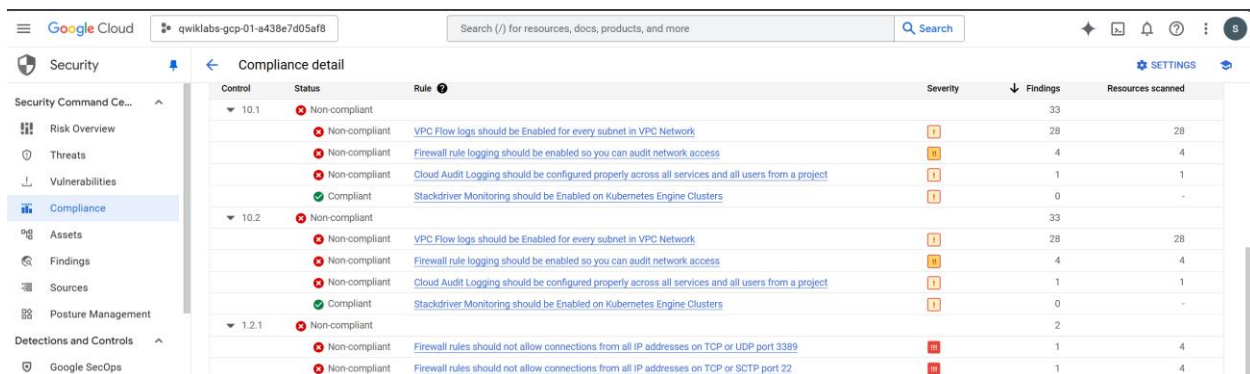


Severity ↓	Finding category	Total findings
!!!	Open RDP port	1
!!!	Open SSH port	1
!!!	Public bucket ACL	1
!!!	Public IP address	1
!!	Firewall rule logging disabled	4
!!	Admin service account	1

Which three resource types are listed with high severity findings?

- ☐ Bucket, Subnetwork, and ServiceAccountKey
- ☐ Network, Firewall, and Bucket
- ☒ Bucket, compute.Instance, and Firewall
- ☐ Network, Subnetwork, and compute.Instance

Submit



The screenshot shows the Google Cloud Security Command Center interface. The left sidebar contains navigation links for Security, Risk Overview, Threats, Vulnerabilities, Compliance (selected), Assets, Findings, Sources, Posture Management, and Google SecOps. The main content area displays a table of compliance findings for the project 'qwiklabs-gcp-01-a438e7d05af8'. The table has columns for Control, Status, Rule, Severity, Findings, and Resources scanned. The findings are grouped by control ID (10.1, 10.2, 1.2.1) and show various non-compliant items with their respective severity levels (High, Medium, Low) and the number of resources affected.

Control	Status	Rule	Severity	Findings	Resources scanned
10.1	Non-compliant	VPC Flow logs should be Enabled for every subnet in VPC Network	High	28	28
	Non-compliant	Firewall rule logging should be enabled so you can audit network access	Medium	4	4
	Non-compliant	Cloud Audit Logging should be configured properly across all services and all users from a project	Low	1	1
	Compliant	Stackdriver Monitoring should be Enabled on Kubernetes Engine Clusters	Low	0	-
10.2	Non-compliant	VPC Flow logs should be Enabled for every subnet in VPC Network	High	28	28
	Non-compliant	Firewall rule logging should be enabled so you can audit network access	Medium	4	4
	Non-compliant	Cloud Audit Logging should be configured properly across all services and all users from a project	Low	1	1
	Compliant	Stackdriver Monitoring should be Enabled on Kubernetes Engine Clusters	Low	0	-
1.2.1	Non-compliant	Firewall rules should not allow connections from all IP addresses on TCP or UDP port 3389	High	1	4
	Non-compliant	Firewall rules should not allow connections from all IP addresses on TCP or SCTP port 22	High	1	4

1) Stop the compromised VM

```
gcloud compute instances stop cc-app-01 \
  --zone=us-central1-c
```

```
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $ gcloud compute instances stop cc-app-01 \
  --zone=us-central1-c
Stopping instance(s) cc-app-01...done.
Updated [https://compute.googleapis.com/compute/v1/projects/qwiklabs-gcp-01-a438e7d05af8/zones/us-central1-c/instances/cc-app-01].
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $
```

2) Create a new disk from the existing snapshot

```
gcloud compute disks create cc-app02-disk \  
  --source-snapshot=cc-app01-snapshot \  
  --zone=us-central1-c
```

```
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $ gcloud compute disks create cc-app02-disk \  
  --source-snapshot=cc-app01-snapshot \  
  --zone=us-central1-c  
Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-01-a438e7d05af8/zones/us-central1-c/disks/cc-app02-disk].  
NAME: cc-app02-disk  
ZONE: us-central1-c  
SIZE GB: 10  
TYPE: pd-standard  
STATUS: READY  
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $
```

3) Create the new VM cc-app-02 using that disk

```
gcloud compute instances create cc-app-02 \  
  --zone=us-central1-c \  
  --machine-type=e2-medium \  
  --disk=name=cc-app02-disk,boot=yes,auto-delete=yes \  
  --no-address \  
  --tags=cc \  
  --service-account=QWIKLABS-USER-SERVICE-ACCOUNT
```

```
DISPLAY NAME: Qwiklabs User Service Account  
EMAIL: qwiklabs-gcp-01-a438e7d05af8@qwiklabs-gcp-01-a438e7d05af8.iam.gserviceaccount.com  
DISABLED: False  
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $ gcloud compute instances create cc-app-02 \  
  --zone=us-central1-c \  
  --machine-type=e2-medium \  
  --disk=name=cc-app02-disk,boot=yes,auto-delete=yes \  
  --no-address \  
  --tags=cc \  
  --service-account=qwiklabs-gcp-01-a438e7d05af8@qwiklabs-gcp-01-a438e7d05af8.iam.gserviceaccount.com  
Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-01-a438e7d05af8/zones/us-central1-c/instances/cc-app-02].  
NAME: cc-app-02  
ZONE: us-central1-c  
MACHINE_TYPE: e2-medium  
PREEMPTIBLE:  
INTERNAL_IP: 10.128.0.4  
EXTERNAL_IP:  
STATUS: RUNNING  
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $
```

4) Stop the new VM to enable Secure Boot

```
gcloud compute instances stop cc-app-02 \
  --zone=us-central1-c
```

```
STATUS: RUNNING
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $ gcloud compute instances stop cc-app-02 \
  --zone=us-central1-c
Stopping instance(s) cc-app-02...done.
Updated [https://compute.googleapis.com/compute/v1/projects/qwiklabs-gcp-01-a438e7d05af8/zones/us-central1-c/instances/cc-app-02].
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $
```

5) Enable Secure Boot on cc-app-02

```
gcloud compute instances update cc-app-02 \
  --zone=us-central1-c \
  --shielded-secure-boot
```

```
Updated [https://compute.googleapis.com/compute/v1/projects/qwiklabs-gcp-01-a438e7d05af8/zones/us-central1-c/instances/cc-app-02].
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $ gcloud compute instances update cc-app-02 \
  --zone=us-central1-c \
  --shielded-secure-boot
Setting shieldedInstanceConfig of instance [cc-app-02]...done.
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $
```

6) Delete the old compromised VM

```
gcloud compute instances delete cc-app-01 \
  --zone=us-central1-c \
  --quiet
```

```
Setting shieldedInstanceConfig of instance [cc-app-02]...done.
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $ gcloud compute instances delete cc-app-01 \
  --zone=us-central1-c \
  --quiet
Deleted [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-01-a438e7d05af8/zones/us-central1-c/instances/cc-app-01].
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $
```

7) Start the new cc-app-02 VM

```
gcloud compute instances start cc-app-02 \
  --zone=us-central1-c
```

```
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $ gcloud compute instances start cc-app-02 \
  --zone=us-central1-c
Starting instance(s) cc-app-02...done.
Updated [https://compute.googleapis.com/compute/v1/projects/qwiklabs-gcp-01-a438e7d05af8/zones/us-central1-c/instances/cc-app-02].
Instance internal IP is 10.128.0.4
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $
```

8) Create a new firewall rule to allow SSH only from 35.235.240.0/20 to the cc tag

```
gcloud compute firewall-rules create limit-ports \
```

```
--network=default \
```

```
--action=ALLOW \
```

```
--direction=INGRESS \
```

```
--rules=tcp:22 \
```

```
--source-ranges=35.235.240.0/20 \
```

```
--target-tags=cc \
```

```
--priority=1000
```

```
Use 'gcloud config set project [PROJECT_ID]' to change to a different project.  
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $ gcloud compute firewall-rules crea  
te limit-ports \  
  --network=default \  
  --action=ALLOW \  
  --direction=INGRESS \  
  --rules=tcp:22 \  
  --source-ranges=35.235.240.0/20 \  
  --target-tags=cc \  
  --priority=1000  
Creating firewall...working..Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-01-a4  
38e7d05af8/global/firewalls/limit-ports].  
Creating firewall...done.  
NAME: limit-ports  
NETWORK: default  
DIRECTION: INGRESS  
PRIORITY: 1000  
ALLOW: tcp:22  
DENY:  
DISABLED: False  
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $
```

9) Create a new firewall rule to allow SSH only from 35.235.240.0/20 to the cc tag

```
gcloud compute firewall-rules update limit-ports \
```

```
--enable-logging
```

```
gcloud compute firewall-rules update default-allow-internal \
--enable-logging
```

10) Delete the default overly broad firewall rules for ICMP, RDP, and SSH

```
gcloud compute firewall-rules delete default-allow-icmp \
--quiet
```

```
gcloud compute firewall-rules delete default-allow-rdp \
--quiet
```

```
gcloud compute firewall-rules delete default-allow-ssh \
--quiet
```

```
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $ gcloud compute firewall-rules delete default-allow-icmp \
--quiet
gcloud compute firewall-rules delete default-allow-rdp \
--quiet
gcloud compute firewall-rules delete default-allow-ssh \
--quiet
Deleted [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-01-a438e7d05af8/global/firewalls/default-allow-icmp].
Deleted [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-01-a438e7d05af8/global/firewalls/default-allow-rdp].
Deleted [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-01-a438e7d05af8/global/firewalls/default-allow-ssh].
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $
```

11) Enable logging on firewall rules


```
gcloud compute firewall-rules update limit-ports \
```

```
--enable-logging
```

```
gcloud compute firewall-rules update default-allow-internal \
```

```
--enable-logging
```

```
Deleted [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-01-a438e7d05af8/global/firewalls/default-allow-internal].
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $ gcloud compute firewall-rules update limit-ports \
--enable-logging

gcloud compute firewall-rules update default-allow-internal \
--enable-logging
Updated [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-01-a438e7d05af8/global/firewalls/limit-ports].
Updated [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-01-a438e7d05af8/global/firewalls/default-allow-internal].
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $
```

11) fix bucket

```
BadRequestException: 400 Invalid bucket name: 'look_bucket_name'
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $ gsutil acl ch -d AllUsers gs://qwiklabs-gcp-01-a438e7d05af8_bucket
Updated ACL on gs://qwiklabs-gcp-01-a438e7d05af8_bucket/
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $
```

```
BadRequestException: 400 Invalid bucket name: 'look_bucket_name'
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $ gsutil acl ch -d AllUsers gs://qwiklabs-gcp-01-a438e7d05af8_bucket
Updated ACL on gs://qwiklabs-gcp-01-a438e7d05af8_bucket/
student_01_8ce02fcc05d1@cloudshell:~ (qwiklabs-gcp-01-a438e7d05af8) $ gsutil pap set enforced gs://qwiklabs-gcp-01-a438e7d05af8_bucket

gsutil uniformbucketlevelaccess set on gs://qwiklabs-gcp-01-a438e7d05af8_bucket
Setting Public Access Prevention enforced for gs://qwiklabs-gcp-01-a438e7d05af8_bucket
Enabling Uniform bucket-level access for gs://qwiklabs-gcp-01-a438e7d05af8_bucket...
```

Lab Complete



Respond and
recover from a data
breach

